

Customer Shopping Behavior Analysis

PROJECT REPORT

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1. Project Overview

This project presents a comprehensive analysis of customer shopping behavior using transactional retail data. The objective was to examine spending patterns, product preferences, customer segments, and subscription behavior in order to generate actionable business intelligence. The analysis was conducted through an end-to-end workflow covering data preparation in Python, structured querying in MySQL, and interactive visualization in Power BI.

2. Dataset Summary

The dataset consists of 3,900 records and 18 columns representing customer transactions across multiple product categories.

Key features include:

- Customer demographics - Age, Gender, Location, Subscription Status
- Purchase details - Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behavior - Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- Missing Data - 37 null values were identified in the Review Rating column

3. Exploratory Data Analysis (Python)

Data preparation and cleaning were performed using Python in a Jupyter Notebook environment. The following steps were carried out:

Data Loading: The dataset was imported using the Pandas library for initial inspection.

Initial Exploration: The structure of the dataset was examined using `df.info()` and summary statistics were generated using `df.describe()` to understand distributions and data types.

Missing Data Handling: Null values in the Review Rating column were imputed using the median rating calculated for each product category, preserving data integrity without distorting overall averages.

Column Standardization: All column names were renamed to snake_case format to ensure consistency and improve readability across the workflow.

Feature Engineering: Two new columns were created to enrich the dataset:

- `age_group` - customers were grouped into age brackets for demographic analysis
- `purchase_frequency_days` - derived from purchase frequency data to measure shopping regularity

Data Consistency Check: The discount_applied and promo_code_used columns were evaluated for redundancy. As both columns carried equivalent information, promo_code_used was dropped to avoid duplication.

Database Integration: The cleaned DataFrame was connected to a MySQL database using SQLAlchemy and PyMySQL, and loaded into a structured table for SQL analysis.

4. Data Analysis using SQL

Ten business queries were executed in MySQL Workbench to extract key insights from the customer dataset:

Q1. Revenue by Gender

Male customers generated a total revenue of \$157,890 compared to \$75,191 from female customers, indicating that male customers are the dominant revenue contributors in this dataset.

Q2. High-Spending Discount Users

A total of 839 customers were identified who applied discounts yet still spent above the average purchase amount, suggesting a strong segment of high-value buyers who are also discount-sensitive.

Q3. Top 5 Products by Average Review Rating

The five highest-rated products based on customer reviews were Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78).

Q4. Shipping Type Comparison

Customers who selected Express shipping had a slightly higher average purchase amount (\$60.48) compared to those who chose Standard shipping (\$58.46), suggesting that higher spenders tend to prefer faster delivery.

Q5. Subscribers vs. Non-Subscribers

The analysis revealed that subscribed customers (1,053) had an average spend of \$59.49 and contributed a total revenue of \$62,645. Non-subscribed customers (2,847) had a slightly higher average spend of \$59.87 with a total revenue of \$170,436. While non-subscribers are more numerous, the average spend per customer remains comparable across both groups.

Q6. Discount-Dependent Products

The five products with the highest percentage of discounted purchases were Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%).

Q7. Customer Segmentation

Customers were classified into three segments based on their purchase history. The majority fell into the Loyal segment with 3,116 customers, followed by Returning customers at 701, and New customers at 83.

Q8. Top 3 Products per Category

Within each product category, the most purchased items were identified. In Accessories, the top products were Jewelry, Sunglasses, and Belt. In Clothing, Blouse, Pants, and Shirt led the rankings. Sandals, Shoes, and Sneakers topped the Footwear category, while Jacket, Coat, and Hoodie were the most purchased in Outerwear.

Q9. Repeat Buyers and Subscription Behavior

Among customers with more than 5 previous purchases, 2,518 were non-subscribers while 958 held active subscriptions, indicating that repeat buying behavior does not strongly correlate with subscription status.

Q10. Revenue by Age Group

The Middle Aged group contributed the highest total revenue at \$88,853, followed by Young Adults at \$78,972 and Seniors at \$65,256.

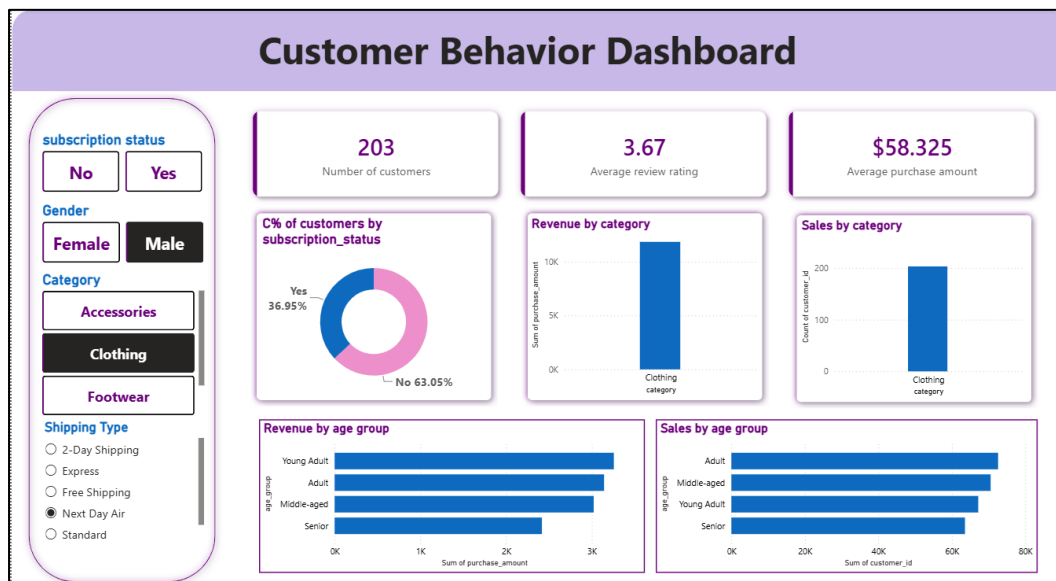
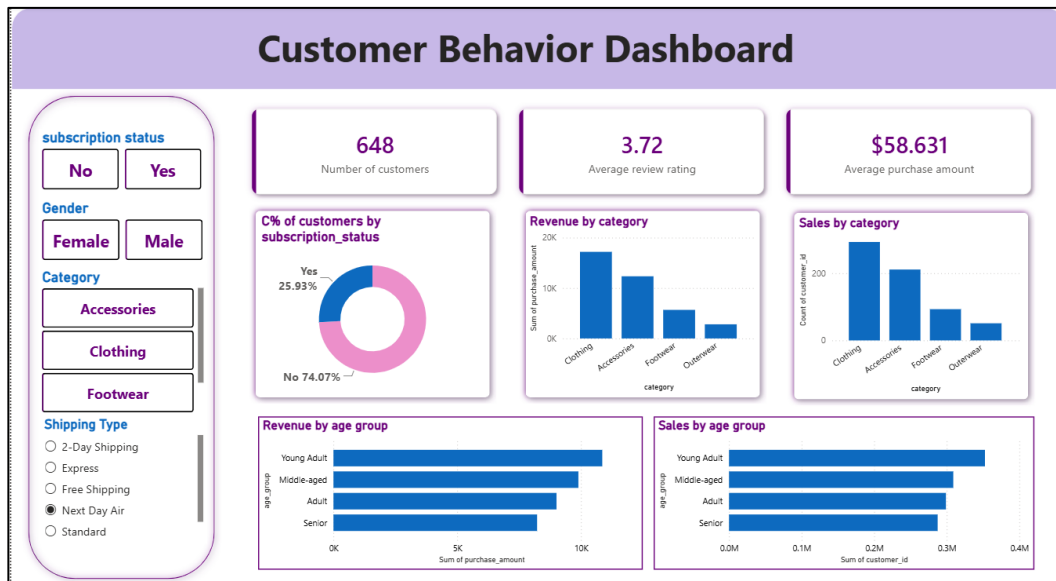
The screenshot displays the MySQL Workbench interface with the 'customer_trends' database selected. The 'Schemas' pane on the left shows the database structure. The 'Query 1' editor contains two SQL queries. The first query, Q4, compares average purchase amounts for Standard and Express shipping. The second query, Q5, compares average spend and total revenue for subscribers and non-subscribers. The 'Result Grid' at the bottom shows the results for Q4.

```
16 GROUP BY item_purchased
17 ORDER BY AVG(review_rating) DESC
18 LIMIT 5;
19
20 -- Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
21 select shipping_type,
22 ROUND(AVG(purchase_amount),2)
23 from customer
24 where shipping_type in ('Standard','Express')
25 group by shipping_type;
26
27 -- Q5. Do subscribed customers spend more? Compare average spend and total revenue
28 -- between subscribers and non-subscribers.
29 SELECT subscription_status,
30 COUNT(customer_id) AS total_customers,
31 ROUND(AVG(purchase_amount),2) AS avg_spend,
32 ROUND(SUM(purchase_amount),2) AS total_revenue
33 FROM customer
34 GROUP BY subscription_status
```

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

5. Dashboard in Power BI

An interactive dashboard was developed in Power BI Desktop by connecting directly to the MySQL database. The dashboard presents key metrics including total number of customers, average purchase amount, and average review rating. Visual components include a donut chart showing customer distribution by subscription status, bar charts comparing revenue and sales across product categories, and horizontal bar charts illustrating revenue and sales performance by age group. Dynamic slicers allow stakeholders to filter data by gender, category, subscription status, and shipping type, enabling flexible and targeted exploration of the data.



6. Business Recommendations

Based on the findings from the SQL analysis and dashboard insights, the following recommendations are proposed:

- **Boost Subscription Uptake** - Since subscribed and non-subscribed customers spend at nearly the same rate, promoting exclusive subscriber benefits such as early access to sales or loyalty rewards could increase subscription conversions without significant cost.
- **Target Male Customers with Premium Campaigns** - As male customers contribute over twice the revenue of female customers, targeted marketing campaigns focusing on male-preferred categories could further maximize revenue.
- **Reward Loyal Customers** - With 3,116 customers already in the Loyal segment, introducing a structured loyalty rewards program would help retain this valuable group and encourage New and Returning customers to progress toward loyalty status.
- **Review Discount Strategy** - Products such as Hat, Sneakers, and Coat show discount dependency rates close to 50%, which may be eroding profit margins. A more strategic discount approach should be considered for these items.
- **Focus on Middle Aged and Young Adult Segments** - These two groups together account for the largest share of revenue. Marketing efforts and product assortment decisions should be aligned with the preferences of these demographics.
- **Leverage Express Shipping Preferences** - Higher spending customers tend to prefer Express shipping. Offering bundled Express shipping deals or free express upgrades above a spending threshold could encourage larger purchases.